



Functional Description

Diesel engine
Series 396TEx4

MS13094/00E

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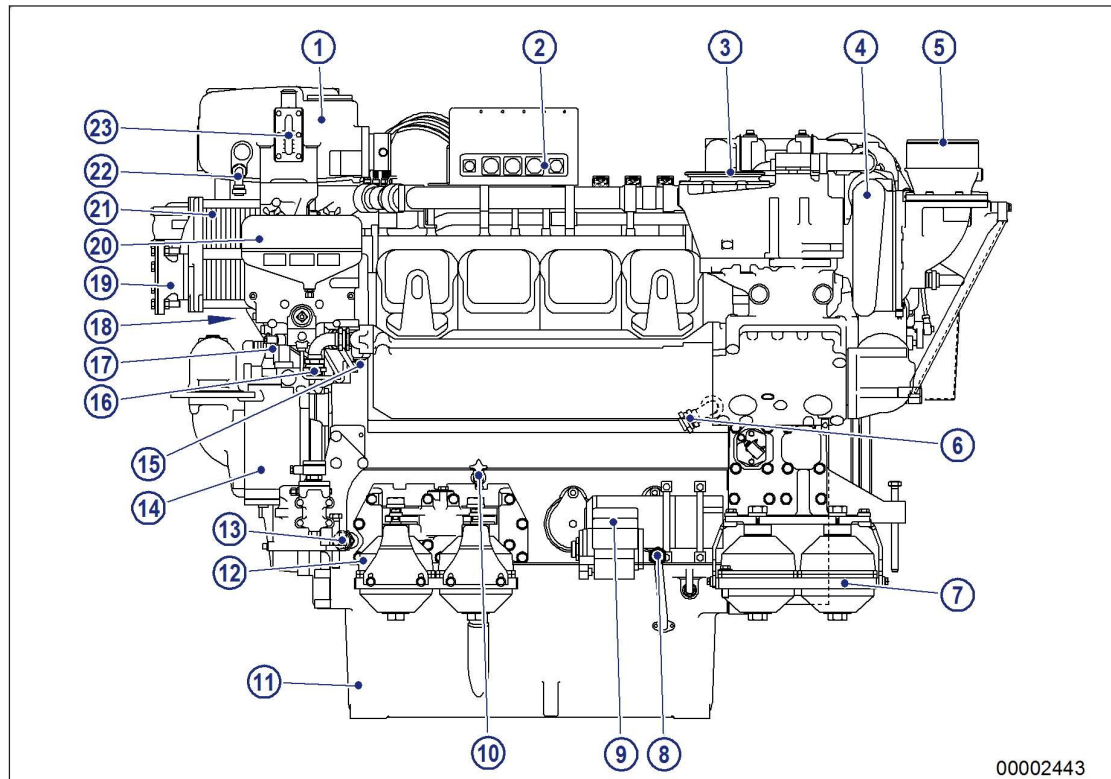
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1 Engines of the Series

1.1 Series V 396 engines

Engine overview 8V LH

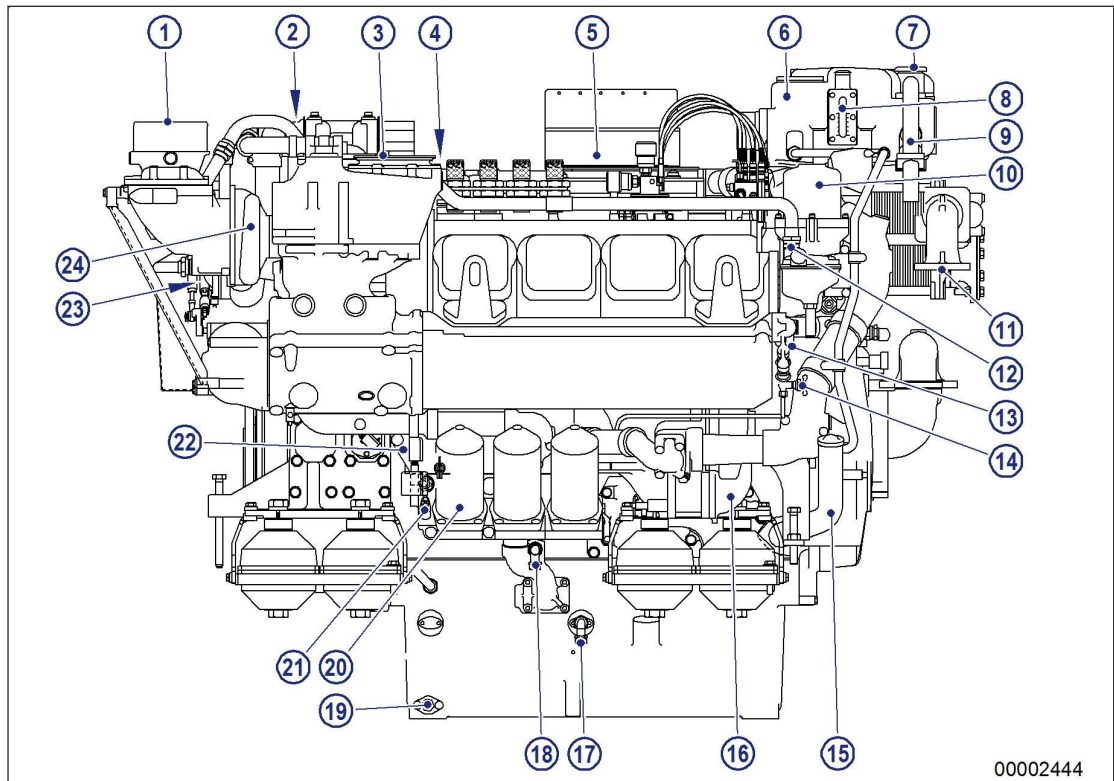


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|---|---|---|
| 1 Engine coolant expansion tank | 9 Air start system (emergency start) | 17 Drain cock on fuel filter |
| 2 Terminal box size III for monitoring | 10 Oil dipstick | 18 Fuel delivery pump |
| 3 Connection, exhaust system | 11 Oil pan | 19 Connection: Seawater inlet from standby pump |
| 4 Turbocharger A | 12 Engine mounting (free end) | 20 Fuel filter |
| 5 Connection: Air inlet before compressor | 13 Connection: Lube oil replenishment | 21 Heat exchanger, raw water/engine coolant |
| 6 Connection: Preheating line, engine inlet | 14 Raw water pump | 22 Drain: Waste water from expansion tank |
| 7 Engine mounting (driving end) | 15 Opening for cleaning, dirt trap | 23 Sight glass: Coolant level in coolant expansion tank |
| 8 Connection: Oil intake | 16 Connection: Preheating line, engine outlet | |

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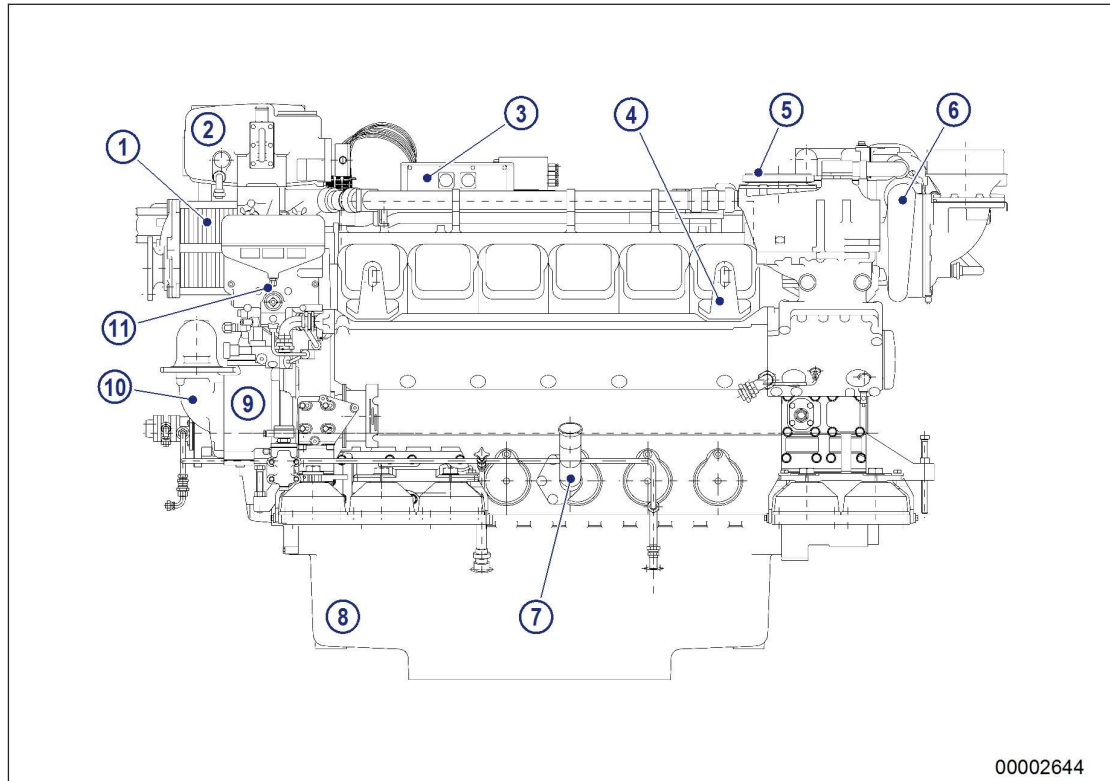
Engine overview 8V RH



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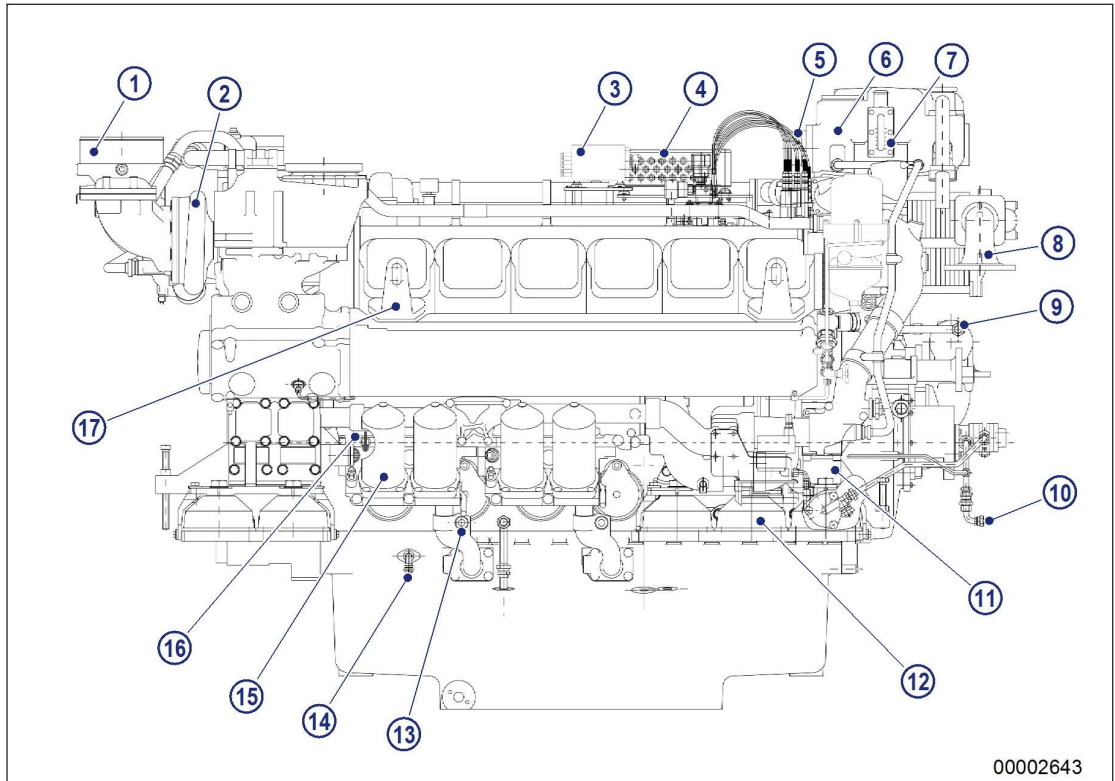
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|--|---|---|
| 1 Connection: Air inlet before compressor | 9 Overflow line | 17 Connection, oil priming pump line outlet |
| 2 Intercooler | 10 Centrifugal oil filter | 18 Connection, oil priming pump line inlet |
| 3 Connection, exhaust system | 11 Connection: Seawater outlet from raw water/engine coolant heat exchanger | 19 Drain plug of oil pan, crankcase lower section |
| 4 Connection: exhaust pyrometer after exhaust turbocharger | 12 Crankcase ventilation | 20 Lube oil filter |
| 5 Terminal box size III for monitoring | 13 Speed sensor for governor | 21 Drain: Contaminants from edge-type filter |
| 6 Engine coolant expansion tank | 14 Connection: Pulse generator for engine speed acquisition | 22 Lube-oil heat exchanger |
| 7 Filler neck for engine coolant | 15 Oil filler neck | 23 Solenoid valve for turbocharger control |
| 8 Sight glass: Coolant level in coolant expansion tank | 16 Engine coolant pump | 24 Turbocharger B |

Engine overview 12V LH



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|---------------------------------|------------------------|-------------------------|
| 1 Plate-core heat exchanger | 5 Exhaust pipework | 9 Raw water pump |
| 2 Engine coolant expansion tank | 6 Exhaust turbocharger | 10 Raw water pump inlet |
| 3 Engine control unit | 7 Oil filler neck | 11 Leak fuel drain |
| 4 Engine lifting eye | 8 Oil pan | |

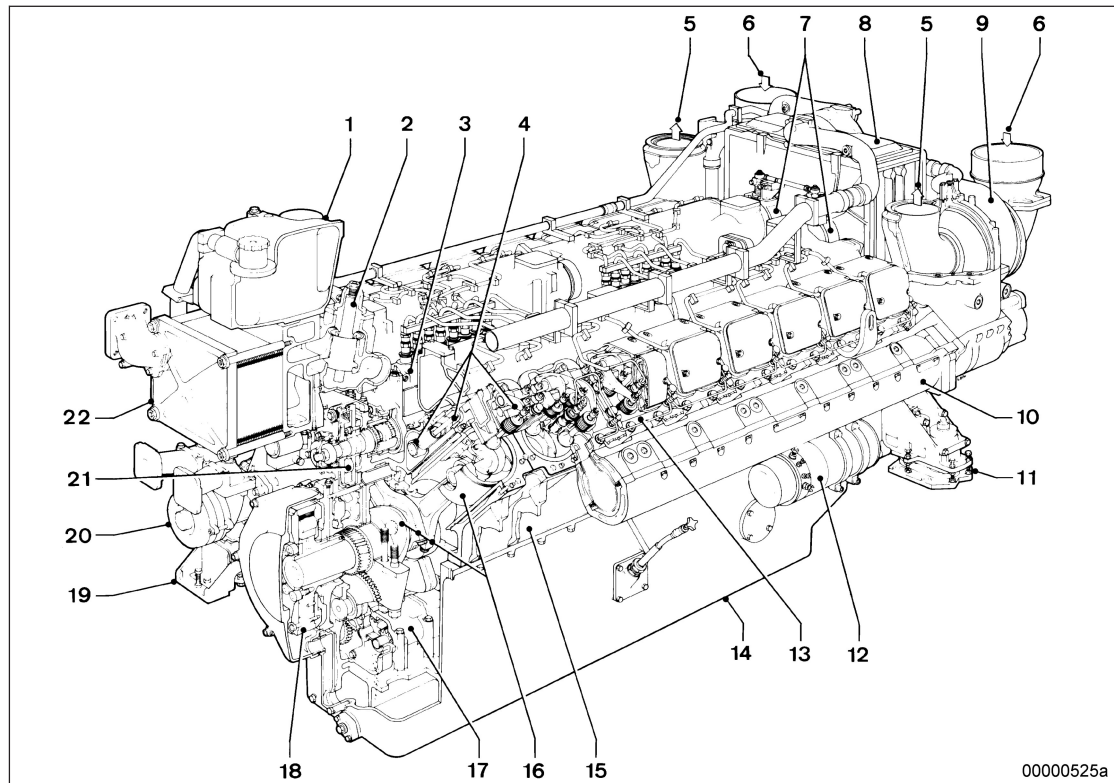
Engine overview 12V RH



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|---|--|----------------------------|
| 1 Intake air inlet | 7 Inspection glass for engine coolant expansion tank | 13 From bypass oil filter |
| 2 Exhaust turbocharger | 8 Raw water outlet from heat exchanger | 14 To bypass oil filter |
| 3 Terminal box assembly | 9 Compressed air supply | 15 Lube oil filter |
| 4 Engine control unit | 10 Lube oil drain | 16 Lube-oil heat exchanger |
| 5 Level monitor for engine coolant expansion tank | 11 Engine coolant pump | 17 Engine lifting eye |
| 6 Engine coolant expansion tank | 12 Engine mounting | |

Engine overview 16 V



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|------------------------------|---------------------------------|------------------------------|
| 1 Coolant expansion tank | 9 Exhaust turbocharger | 17 Oil pump |
| 2 Coolant thermostat | 10 Exhaust manifold | 18 Vibration damper |
| 3 Injection pump | 11 Engine mounting, driving end | 19 Engine mounting, free end |
| 4 Engine control unit | 12 Starter | 20 Bilge pump |
| 5 Exhaust outlet | 13 Cylinder head | 21 Injection timer |
| 6 Air inlet | 14 Oil pan | 22 Engine coolant cooler |
| 7 Emergency air-shutoff flap | 15 Crankcase | |
| 8 Intercooler | 16 Crank drive | |

Series V 396 engines

The engine series covers a wide range of possible applications.

These engines are compact, reliable, easy to service as well as economical.

They are also characterized by low exhaust gas and noise emissions.

With their sequential turbocharging, electronic engine management and split-circuit cooling system these engines are state-of-the art technology.

Technical data

- Four-stroke, four valves per cylinder and direct injection
- 8 cylinders
- 12 cylinders
- 16 cylinders
- 90° V-angle
- Power:
 - 8V = 1000 kW
 - 12V = 1200 kW
 - 16V = 2000 kW
 - 125 kW per cylinder
- Displacement:
 - 8V = 31.68 l
 - 12V = 47.5 l
 - 16V = 63.36 l
 - 3.96 l per cylinder
- Speed:
 - 1900 rpm
- Counterclockwise rotation
- Electronically controlled fuel injection
- Sequential turbocharging with charge-air cooling
- Triple-walled, water-cooled exhaust lines
- Electric or pneumatic starter
- Resilient, height-adjustable engine mounting
- Service block with pumps, filters, heat exchangers, battery-charging generator, coolant expansion tank etc.
- Electronic engine management system

Benefits

Life cycle costs

- Low fuel consumption over entire performance map
- Low oil consumption
- Easy maintenance
- Long service intervals

Drivability

- Unrestricted low-load operation
- High power

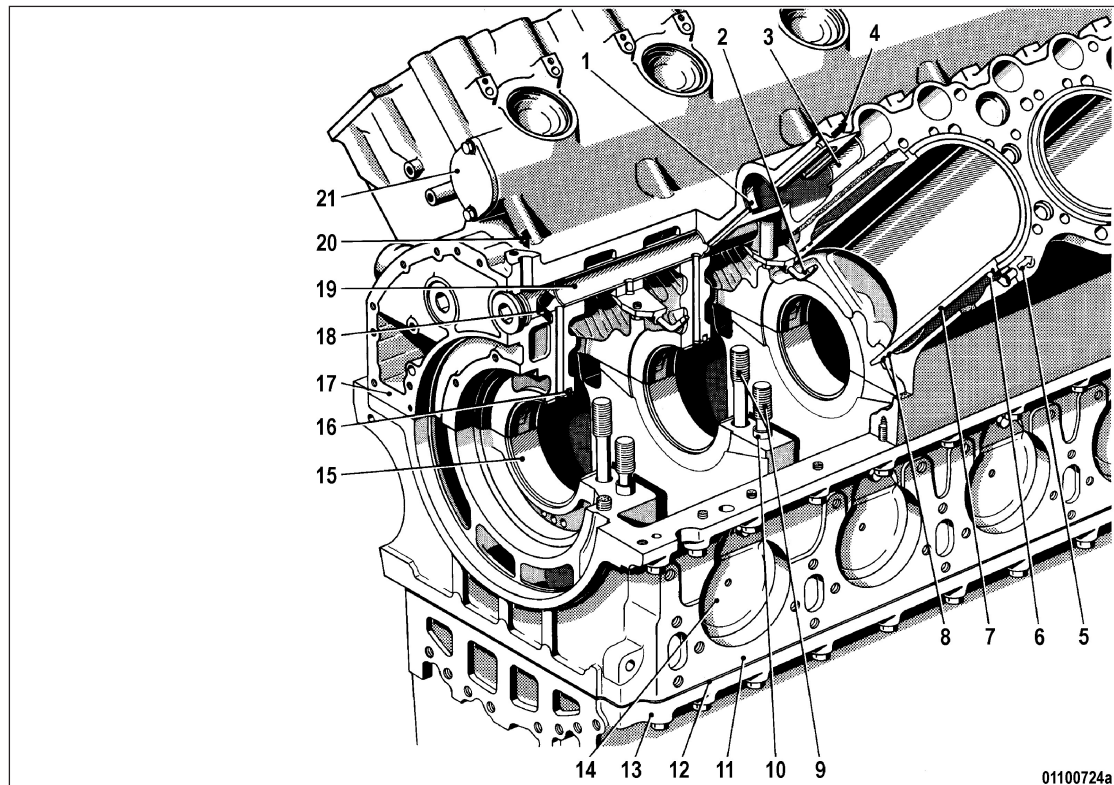
Environmental compatibility

- IMO-NOx certified
- Low fuel consumption

Safety

- High standard
- Compliance with DIN/ISO requirements

1.2 Crankcase



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|-----------------------------------|-------------------------|-------------------------------------|
| 1 Camshaft bearing | 8 Sealing ring | 15 Crankshaft bearing |
| 2 Oil spray nozzle | 9 Bearing cap stud | 16 Alignment pin |
| 3 Tappet guide bushing | 10 Sleeve | 17 Crankcase upper deck |
| 4 Engine oil to cylinder head | 11 Crankcase lower deck | 18 Engine oil inlet |
| 5 Engine coolant to cylinder head | 12 Gasket | 19 Main oil gallery |
| 6 O-ring | 13 Oil pan | 20 Engine oil to distribution block |
| 7 Cylinder liner | 14 Cover | 21 Camshaft cover plate |

Crankcase

The crankcase is the main structural component of the engine and includes an upper and a lower deck. It accommodates the running gear in bearings and provides the attaching points for engine accessories and engine mounts.

The upper deck of the crankcase is fitted with cast-in transverse partitioning walls ensuring a compact design and high torsional rigidity. The cylinder banks are arranged at an angle of 90°.

The lower deck of the crankcase is secured to the lower side of the upper deck with screws. The transverse partition walls of the upper deck represent the crankshaft bearing caps.

The oil pan seals the crank chamber at the bottom.

Gearcase and flywheel housing are mounted on the crankcase.

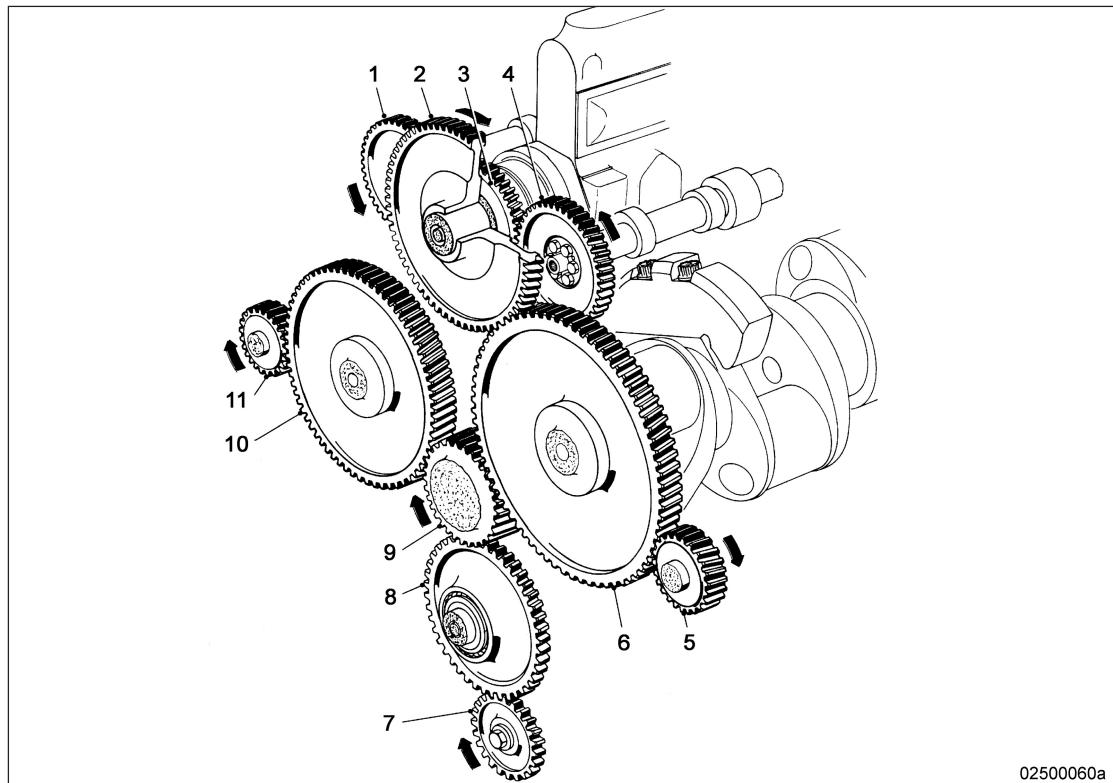
Technical Data

- Two-part cast crankcase
- Integral coolant ducting
- One integral main oil gallery
- Replaceable, wet cylinder liners
- Split sleeve bearings for crankshaft
- Sleeve bearings for camshaft
- Piston cooling-oil spray nozzles in crankshaft upper deck
- Crankcase ventilation (closed circuit)
- Access ports in the crankcase lower part

Benefits

- High rigidity
- Low noise and vibration levels
- Easy maintenance

1.3 Gear train



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|-----------------------------------|---------------------------------------|--|
| 1 Camshaft drive gear, right side | 5 Drive gear of raw water pump | 9 Crankshaft gear |
| 2 Drive gear for injection pump | 6 Large idler gear, left side | 10 Coolant pump idler gear, right side |
| 3 Camshaft idler gear | 7 Drive gear of engine oil pump | 11 Drive gear of coolant pump |
| 4 Camshaft drive gear, left side | 8 Idler gear of engine oil pump drive | |

Free end (KGS)

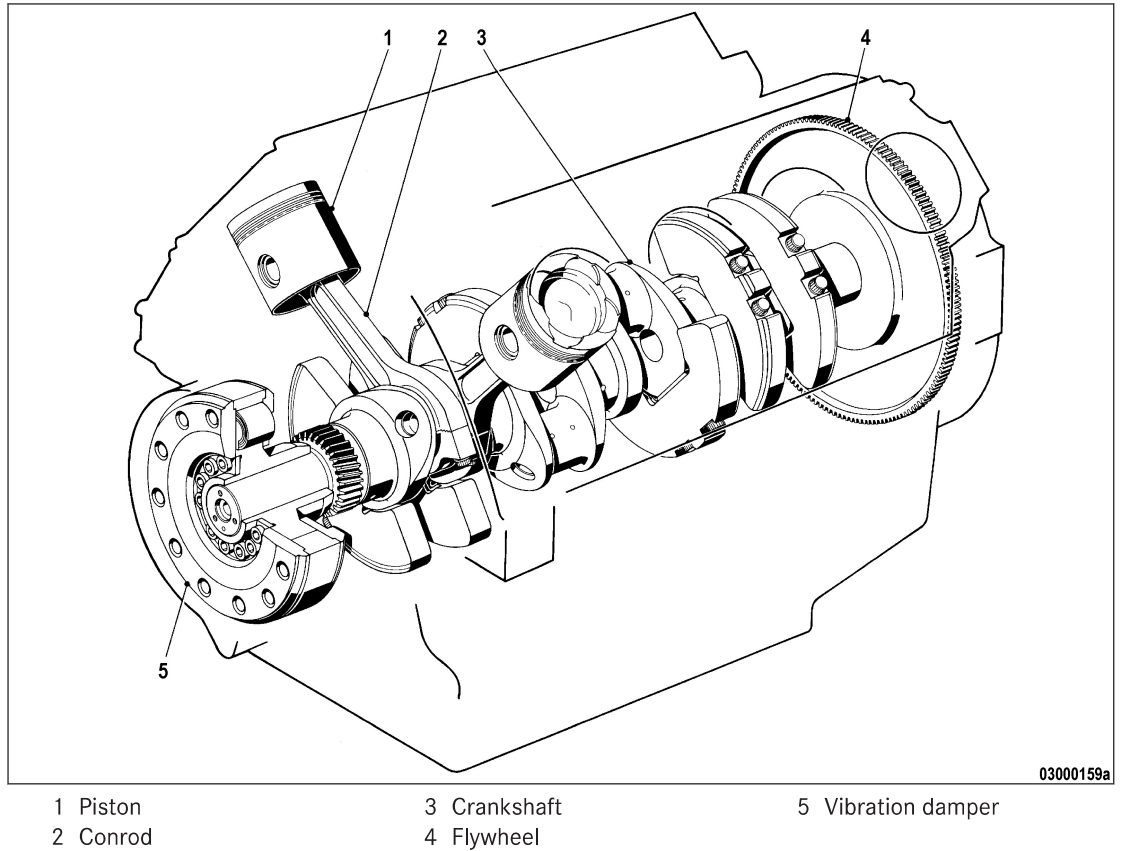
The crankshaft gearwheel drives the following assemblies via idler gears:

- Coolant pump
- Raw water pump
- Fuel injection pump
- Engine oil pump
- Camshafts

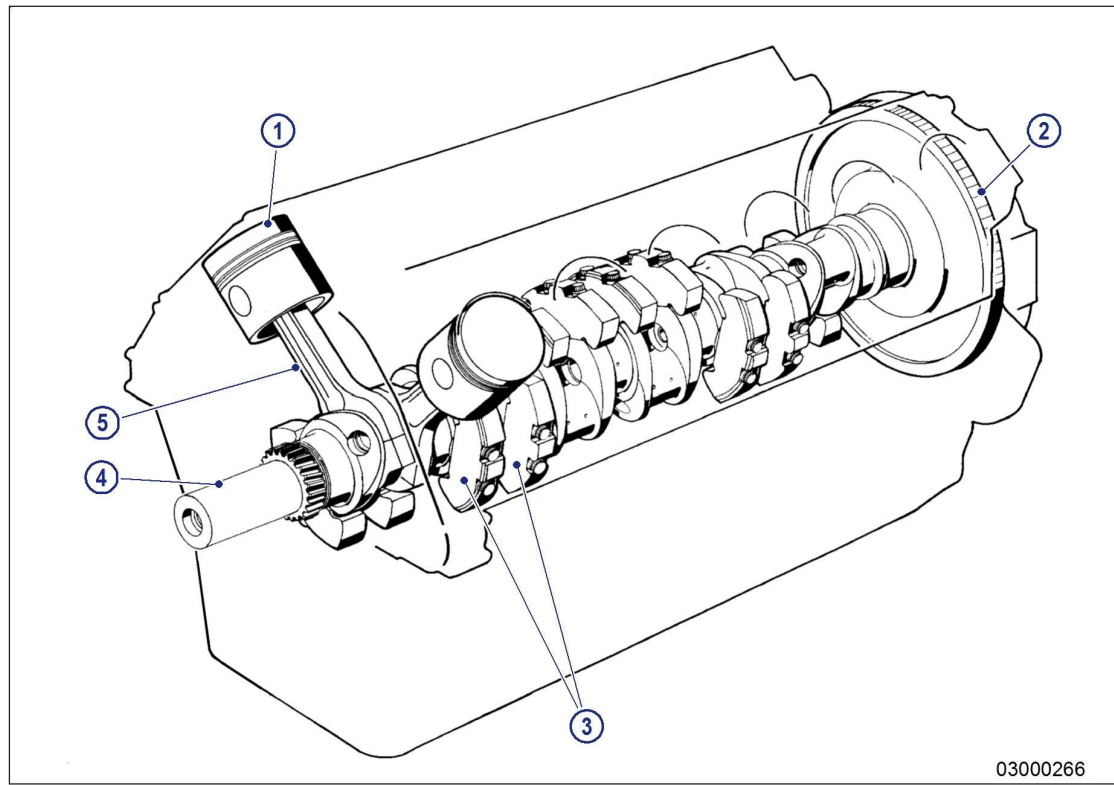
Drive gears and idler gears are located in the gearcase, which is mounted on the crankcase.

1.4 Crank drive

Crank drive of 8V engines



Crank drive of 12V engines



1 Piston
2 Flywheel

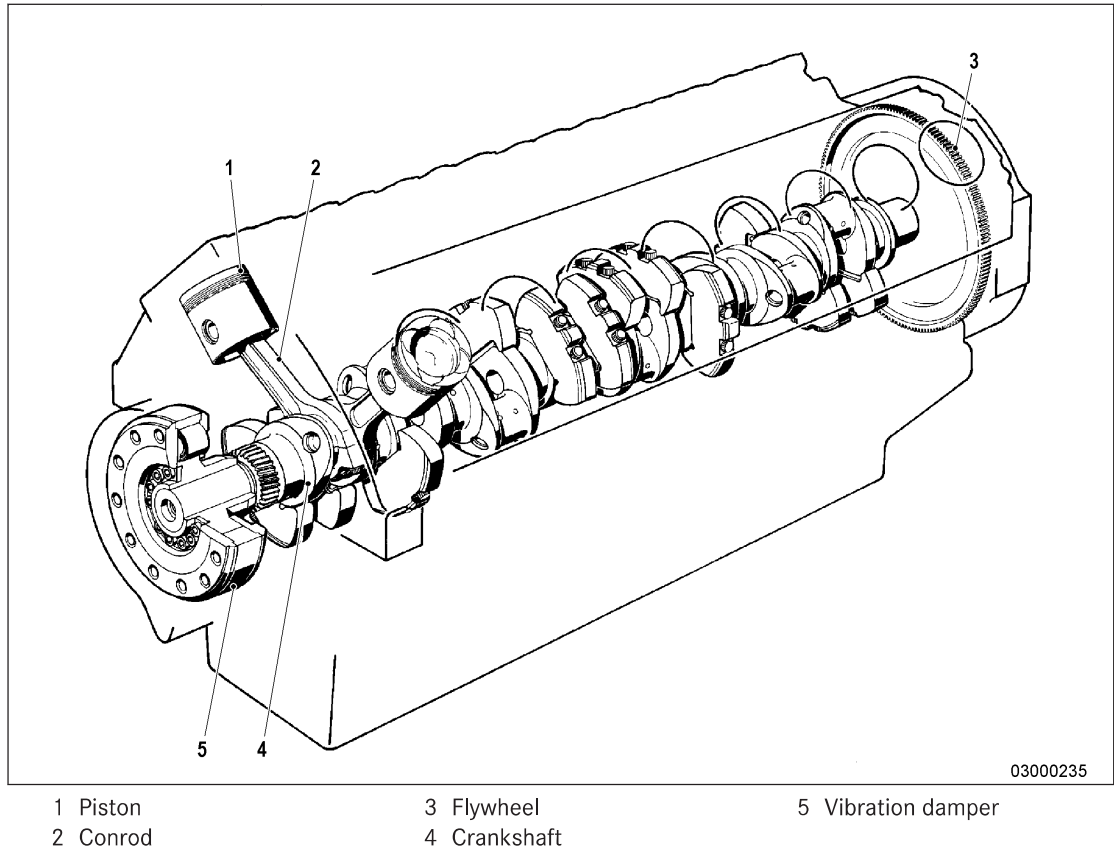
3 Counterweights
4 Crankshaft

5 Conrod

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Crank drive of 16V engines



Technical data

Forces generated by combustion in the cylinder chambers are transmitted to the crankshaft via the pistons and conrods. This sets the crankshaft turning whereby the resultant rotational movement is transferred to the driven equipment via the flywheel and the steel-spring coupling attached to it. Carefully matched components ensure maximum performance and minimum wear.

Piston

- Light metal skirt with bolt-on steel piston crown
- 2 compression rings, 1 oil control ring
- Piston cooling via oil spray nozzles

Conrod

- Forged
- Low-wear plain bearing in conrod eye and conrod base
- Split bearing shells in conrod base
- Lubrication of bearing in conrod eye via conrod oil gallery
- Oil supply to bearing in conrod foot via crankshaft

Crankshaft

- Forged
- Bolt-on counterweights
- Low-wear sleeve bearings, oil supply from lube oil system